

Online Algorithms Fundamentals

1 Core Idea

Online Algorithm

An **online algorithm** receives input requests one by one. After each request arrives, it must make a decision before seeing future requests.

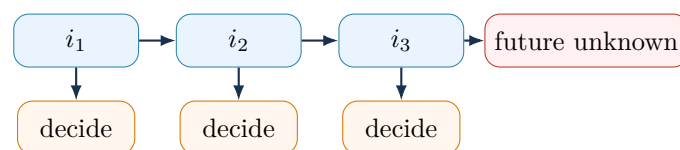
Offline Algorithm

An **offline algorithm** sees the whole input sequence in advance. It can use future information to make the best possible decisions.

Main Difference

Online algorithms make decisions with incomplete information. Offline algorithms make decisions with complete information.

2 Decision Timeline



After request i_j , the algorithm knows only

$$i_1, i_2, \dots, i_j.$$

It does not know

$$i_{j+1}, i_{j+2}, \dots$$

3 Examples of Online Problems

Problem	Online decision	Why future matters
Ski rental	Rent today or buy skis.	The season length is unknown.
Caching	Choose which page to evict.	Future page requests are unknown.
Scheduling	Assign a job to a machine.	Future jobs may be larger or smaller.
Routing	Move a server or route a request.	Future request locations are unknown.

4 Cost Notation

Let I be one complete input sequence.

$C_A(I)$ = cost paid by online algorithm A on I .

$\text{OPT}(I)$ = minimum possible offline cost on the same I .

Fair Comparison

Both algorithms are tested on the same input I . The online algorithm sees I gradually, while OPT sees all of I before deciding.

5 Competitive Ratio

Cost-Minimization Ratio

For a minimization problem,

$$\text{competitive ratio of } A = \max_I \frac{C_A(I)}{\text{OPT}(I)}.$$

The maximum is over all possible input sequences. If

$$C_A(I) \leq r \cdot \text{OPT}(I)$$

for every input I , then A is called **r -competitive**.

How to Read r

$r = 1$ means optimal, $r = 2$ means at most twice the offline optimum.

6 Additive Constant

Sometimes the definition allows a fixed constant b :

$$C_A(I) \leq r \cdot \text{OPT}(I) + b.$$

The constant b handles very small inputs where a fixed startup cost can distort the ratio. For most introductory calculations, use the simpler ratio

$$\frac{C_A(I)}{\text{OPT}(I)}.$$

7 Worked Example

Problem

For three input sequences, an online algorithm A and the offline optimum have these costs:

I	I_1	I_2	I_3
$C_A(I)$	12	15	10
$\text{OPT}(I)$	8	10	5

Find the worst observed competitive ratio.

Solution

Compute the ratio for each input:

$$\frac{C_A(I_1)}{\text{OPT}(I_1)} = \frac{12}{8} = \frac{3}{2}.$$

$$\frac{C_A(I_2)}{\text{OPT}(I_2)} = \frac{15}{10} = \frac{3}{2}.$$

$$\frac{C_A(I_3)}{\text{OPT}(I_3)} = \frac{10}{5} = 2.$$

The worst observed ratio is the largest one:

$$\max \left\{ \frac{3}{2}, \frac{3}{2}, 2 \right\} = 2.$$

Answer

Worst observed ratio = 2.

8 How to Prove an Algorithm Is Competitive

Proof Template

To prove that an online algorithm A is r -competitive, show that every input I satisfies

$$C_A(I) \leq r \cdot \text{OPT}(I).$$

A typical proof has three parts:

1. Describe the cost paid by the online algorithm.
2. Give a lower bound on $\text{OPT}(I)$.
3. Compare the two expressions and show the ratio is at most r .

Important

Do not compare the online algorithm to a weak offline strategy. Compare it to OPT , the best possible offline solution.

9 Minimization vs Maximization

For cost problems, smaller is better, so use

$$\frac{C_A(I)}{\text{OPT}(I)}.$$

For profit problems, larger is better, so the ratio is usually written in the opposite direction:

$$\frac{\text{OPT}(I)}{C_A(I)}.$$

Use the Correct Direction

Cost problem:

$$C_A(I) \leq r \text{OPT}(I).$$

Profit problem:

$$\text{OPT}(I) \leq r C_A(I).$$

10 Deterministic and Randomized Online Algorithms

Type	Meaning
Deterministic online algorithm	Makes the same decision whenever it sees the same history.
Randomized online algorithm	Uses random choices while making online decisions.

For randomized algorithms, we usually compare the **expected cost** to the offline optimum:

$$\mathbb{E}[C_A(I)] \leq r \cdot \text{OPT}(I).$$

11 Adversarial Inputs

Competitive analysis is worst-case. This means the input sequence may be chosen to make the online algorithm look bad.

Why Worst Case?

Online algorithms cannot rely on the future being friendly. Competitive analysis asks how well the algorithm performs even on difficult input sequences.

12 Review Box

Core Facts

Online algorithm: sees requests one at a time.

Offline optimum: sees the whole input first.

$C_A(I)$ = online cost on input I .

$\text{OPT}(I)$ = best offline cost on I .

For minimization: A is r -competitive if $C_A(I) \leq r \text{OPT}(I)$.

Competitive ratio measures worst-case loss from not knowing the future.